

# The Determination of Finger-Flexor Critical Force in Rock Climbers

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**Purpose:** To determine if the mathematical model used for the estimation of critical force (CF) and the energy store component  $W'$  are applicable to intermittent isometric muscle actions of the finger flexors of rock climbers, using a multisession test. As a secondary aim, the agreement of estimates of CF and  $W'$  from a single-session test was also determined. The CF was defined as the slope coefficient, and  $W'$  was the intercept of the linear relationship between total "isometric work" ( $W_{lim}$ ) and time to exhaustion ( $T_{lim}$ ). **Methods:** Subjects performed 3 (separated by either 20 min or >24 h) tests to failure using intermittent isometric finger-flexor contractions at 45%, 60%, and 80% of their maximum voluntary contraction. **Results:** Force plotted against  $T_{lim}$  displayed a hyperbolic relationship; correlation coefficients of the parameter estimates from the work-time CF model were consistently very high ( $R^2 > .94$ ). Climbers' mean CF was 425.7 (82.8) N (41.0% [6.2%] maximum voluntary contraction) and  $W'$  was 30,882 (11,820) N·s. Good agreement was found between the single-session and multisession protocol for CF (intraclass correlation coefficient [ICC<sub>3,1</sub>] = .900; 95% confidence interval, .616–.979), but not for  $W'$  (ICC<sub>3,1</sub> = .768; 95% confidence interval, .190–.949). **Conclusions:** The results demonstrated the sensitivity of a simple test for the determination of CF and  $W'$ , using equipment readily available in most climbing gyms. Although further work is still necessary, the test of CF described is of value for understanding exercise tolerance and to determine optimal training prescription to monitor improvements in the performance of the finger flexors.

**Keywords:** rock climbing, exercise testing, anaerobic capacity, power-duration relationship

Rock climbing requires repeated isometric contractions of the finger flexors, which are responsible for flexion of the metacarpophalangeal and interphalangeal joints.<sup>1</sup> These contractions cause regular periods of ischemia in the forearms; the extent of this ischemia and the subsequent recovery from it have been shown to differentiate ability groups of rock climbers<sup>2</sup> and disciplines<sup>3</sup> and are likely to be a trainable characteristic.<sup>4</sup> As such, the fatigue resistance of the finger flexors is considered one of the most important factors in climbing performance. However, although methods for the determination of maximal finger flexor strength have been described in the literature,<sup>5</sup> as yet, there are no tests to determine functional aerobic metabolic capacity, delineating steady and nonsteady states in rock climbers.

During high-intensity muscular exercise, the time for which exercise can be sustained decreases as a hyperbolic function of increasing power, speed, tension, or force (eg, power illustrated in Figure 1).<sup>6</sup> Consequently, performance and the point of exhaustion are highly predictable. When work data are plotted against time, it may be observed that power output falls as a function of the duration of exercise and that it levels off (asymptotes on the abscissa). The point of leveling off is termed as critical power (CP) and is defined as the maximum work that a muscle group can maintain for an extended duration without fatigue, whereas the work capacity that may be completed above CP is termed as  $W'$  (often described as the "energy store" component).<sup>6</sup> While CP is limited by the availability of oxidative substrates (glycogen), hyperthermia, and central fatigue,  $W'$  is limited by progressive depletion of high-energy phosphates and accumulation of metabolites associated with peripheral fatigue.<sup>7</sup>

Despite the majority of research exploring isotonic muscle actions, the same relationship is also true for isometric muscle action, despite no mechanical work being done.<sup>8,9</sup> Previous research investigating isometric work has utilized an analog of mechanical work termed as limit work ( $W_{lim}$ , in newton seconds), which is calculated as the force ( $F$ , in newtons) of the isometric muscle action multiplied by the time to exhaustion (or limit time,  $T_{lim}$ , in seconds) that the  $F$  can be maintained. As this paper is concerned with finger flexor force, the isometric analogs' critical force (CF, in newtons) and  $W'$  (in newton seconds) will be referred to (eg, Hendrix et al<sup>8</sup>).

Methods for the determination of CF have been demonstrated for a number of synergistic muscle groups.<sup>9,10</sup> However, there is a paucity of data describing the use of CF tests in climbers, specifically for the finger flexors. Although the importance of fatigue resistance of the finger flexors in climbers has been investigated,<sup>2,11</sup> this is the first study to utilize a threshold test applicable to climbing populations. Kellawan and Tschakovsky<sup>10</sup> have previously described a methodology for the determination of CF in the forearms, using grip dynamometry. However, given the lack of specificity of grip dynamometry to climbing performance,<sup>12,13</sup> there is a need for ecologically validated tests within the sport. Determining finger flexor CF would be advantageous in understanding exercise tolerance in climbers, and determining optimal training prescription and monitoring. Therefore, the present study aims to determine if the mathematical model used for estimating CF and  $W'$  is applicable to intermittent isometric muscle actions of the finger flexors of rock climbers, using a multisession test. As a secondary aim, the agreement of estimates of CF and  $W'$  obtained in a single session with 20 minutes of recovery will be compared with those obtained from the multisession test (>24 h of recovery).

## Methods

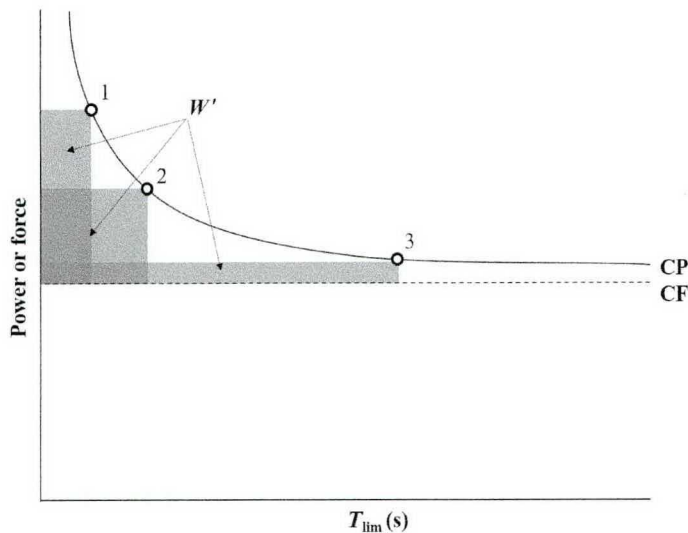
### Subjects

In total, 11 healthy male rock climbers (mean [SD]: age = 27 [6] y, height = 1.76 [0.06] m, and body mass = 69.2 [4.7] kg)

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volunteered to participate. Subjects were included based on having a minimum of 3 years of climbing experience and no known cardiovascular or respiratory diseases or illnesses. Subjects were familiar with climbing-specific forearm training and exhaustive forearm exercise and were free from injury. Using self-reported ability,<sup>14</sup> the subjects were categorized as advanced to elite level climbers (maximum 6-mo red-point grade of French 7b – 8b+; Union Internationale des Associations d'Alpinisme [UIAA] IX- – X+). The subjects gave written informed consent to participate in this



**Figure 1** — Illustration of the power or force–time relationship for high-intensity exercise. The numbered points (1–3) represent time to exhaustion for independent tests at the power or force designated for each. The hyperbolic relationship is defined by 2 parameters: the asymptote for power or force and the curvature constant  $W'$  (represented by the rectangular boxes above CP/CF and expressed in kilojoules or newton seconds, respectively). The CP/CF defines the upper boundary of the heavy-intensity domain and represents the highest power sustainable without drawing continuously upon  $W'$ . Severe-intensity exercise, above CP/CF, results in exhaustion when  $W'$  has been expended. CP indicates critical power; CF, critical force;  $W'$ , energy store component;  $T_{lim}$ , time limit.

study, which was approved by the University of Derby Human Sciences Research Ethics Committee (17-1718DGs), conforming to the Code of Ethics of the World Medical Association (Declaration of Helsinki).

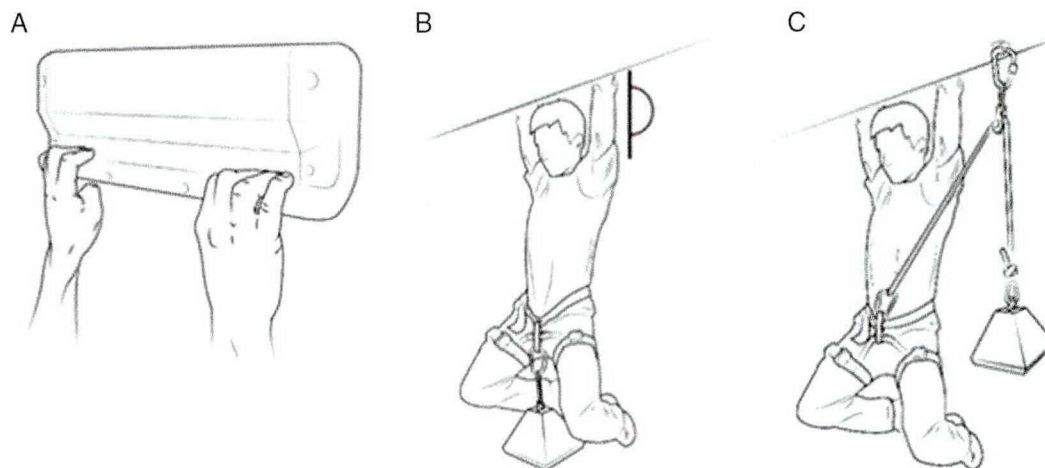
## Design

Subjects were required to visit the laboratory on 4 occasions. During the first visit, they completed a standardized warm-up, were familiarized with the equipment and testing protocol, and completed the finger flexor maximum voluntary contraction (MVC) test as well as intermittent submaximal isometric contractions requiring 80%, 60%, and 45% of MVC to exhaustion, with a work to relief ratio of 7:3 seconds and 20 minutes of recovery between. The subsequent 3 visits were randomized in order. During each visit, subjects performed the same standardized warm-up followed by one of the 3 separate series of intermittent isometric contractions to failure at 80%, 60%, or 45% of MVC. Force and  $T_{lim}$  were recorded on each occasion.

Subjects attended the laboratory at the same time of day ( $\pm 2$  h) and in a rested state (having performed no heavy exercise in the 24 h preceding the test) having refrained from consuming food and caffeinated beverages for 3 hours prior. Before each session, subjects completed a standardized warm-up consisting of 5-minute pulse-raising activity; mobilizing (walking, jogging skipping, etc); 5 minutes of climbing; and 4 sets of 40 seconds of 7:3 hangs on the testing rung in a half-crimp position (Figure 2A). Subjects attended 4 testing sessions, which were completed within a 2-week period; visits were separated by a minimum of 24 hours.

## Methodology

**Testing Rung and Hand Positioning.** During all visits, each finger hang was performed on a Lattice Training (Sheffield, England) rung (20 mm deep and 10 mm radius; Figure 2A). All tests were performed in a 2-handed half-crimp hang position (90° flexion at the proximal interphalangeal joint with the thumb not engaged in the grip). In accordance with Baláš et al,<sup>13</sup> subjects were instructed to hang with arms extended above the head (180° shoulder flexion), maintaining a slight bend in the elbow with shoulders engaged (Figure 2B). To modify the load, weight



**Figure 2** — (A) “Half-crimp” position, 90° flexion at the proximal interphalangeal joint with the thumb not engaged in the grip; (B) climber performing 2-handed hang on lattice rung, with slight bend in arms and engaged shoulders with additional weight; and (C) with assistance. Note: Pulley and weight were located directly in front of the subject; this is not shown in the illustration. Illustrations reproduced with permission from lattoxentraining images.



increments from 0.5 to 5 kg were either added or removed using a pulley system attached to a climbing harness worn by the subject.

**Determination of MVC.** Finger flexor MVC was determined during the first visit by performing a 2-handed half-crimp hang. The MVC was defined by the maximum weight held for 7 seconds while maintaining a half-crimp position. Subjects were allowed up to 6 attempts; weight was added to the subject's body mass until finger flexor and extensor MVC was achieved. A 2-minute rest was provided between each MVC attempt.

**Determination of CF.** The determination of CF was based on the methodology of Monod and Scherrer,<sup>9</sup> and more recently Hendrix et al.,<sup>8</sup> involving fatiguing muscle actions of the finger flexors at 3 different intensities. However, instead of continuous fatiguing isometric exercise used in these previous studies, subjects performed intermittent isometric exercises at a 7:3 work-to-rest cycle. Intermittent, but not continuous, time to failure was chosen as it has previously been shown to differentiate between climbers and nonclimbers.<sup>1,15</sup> During the "work" phase, subjects were instructed to maintain a half-crimp position while hanging at a predetermined percentage of their body mass (%MVC). During the "rest" phase, to standardize practice, subjects were instructed to be in the anatomical position, during which they could apply climbing chalk but not shake their forearms or hands (shaking of the hands is known to aid recovery, eg, Baláš et al<sup>16</sup>). The goal was to achieve time to exhaustion ( $T_{lim}$ ) between 1 and 15 minutes, within which the time to failure should be hyperbolic.<sup>7</sup> Pilot work suggested that the CF lay below 45% MVC for most climbers; thus, to ensure that fatigue would occur in less than 20 minutes, work was performed at 80%, 60%, and 45% of MVC. Testing was halted if the subject exceeded 20 minutes. As previously described, weight was either added or subtracted from the subject's body mass using a harness and pulley system to achieve the desired percentage of MVC.

Exhaustion was defined as failure to complete a hang or failing to maintain the hold in the half-crimp position, determined visually by the experimenter. Visual and audio cues were given via an electronic device to signal the work and rest intervals. Subjects were blind to the relative intensity of each trial (percentage of MVC) but were informed that the task was fatiguing, that they should keep going until they were no longer able to hold the rung, and that the accuracy of the test was reliant on continuing to failure. Elapsed time and clocks were hidden from the subject's sight. Subjects were given verbal encouragement to reach task failure.

The  $W_{lim}$  for the intermittent isometric muscle actions was calculated by multiplying the  $F$  of the muscle actions by the  $T_{lim}$ . The CF was determined from the 3 submaximal tests (multisession protocol and single-session protocol). Linear regression was used to provide 2 sets of CF and  $W'$  estimates from the results of these trials, using the work-time (Equation 1) and the 1/time models (Equation 2), see Figures 3B and 3C. The work-time model plots  $W_{lim}$  against  $T_{lim}$ ;  $W'$  is given by the y-intercept and CF as the slope of Equation 1. The 1/time model plots force against  $1/T_{lim}$ ; CF is given by the y-intercept and  $W'$  as the slope of Equation 2.

$$W_{lim} = T_{lim} \cdot CF + W' \quad (1)$$

$$F = W' \cdot \left( \frac{1}{T_{lim}} \right) + CF \quad (2)$$

where  $W_{lim}$  is the work limit (in newton seconds),  $T_{lim}$  is the time limit (in seconds), CF is the critical force (in newtons),  $W'$  is the energy store component (in newton seconds),  $F$  is the force (in newtons).

## Statistical Analysis

Normal distributions were ascertained, and homogeneity of variances was confirmed after visual assessment of the frequency histogram and a Shapiro-Wilk test, respectively. All values are reported as mean (SD). Correlation coefficients were calculated for CF and  $W'$  derived from the work-time and 1/time models. The agreement between the values of CF and  $W'$  obtained from the single-session test was compared with the multisession test by calculating the intraclass correlation coefficient (model 3.1).<sup>17</sup> Bland-Altman plots were constructed for CF and  $W'$ , and 95% limits of agreement were calculated for both. Analysis was conducted using the SPSS statistical software package (release 24, 2016; IBM SPSS statistics; SPSS Inc, Chicago, IL).

## Results

### 3-Parameter Multisession CF Protocol

All subjects' data from the multisession CF protocol demonstrated a hyperbolic relationship between force (y-axis; in newtons) and  $T_{lim}$  (x-axis; in seconds), where the CF is indicated by the force asymptote and the  $W'$  is the curvature constant (Figure 3A). Figure 3 also demonstrates the derivation of the parameter estimates using the work-time (Figure 3B) and 1/time models (Figure 3C). Although there were no significant differences in the parameter estimates from the different CF models and the correlation coefficients were consistently very high for both models (work-time model range:  $R^2 = .94$ – $1.00$ ; 1/time model range:  $R^2 = .86$ – $1.00$ ), the work-time model generally fit the data better and was therefore used for subsequent analysis.

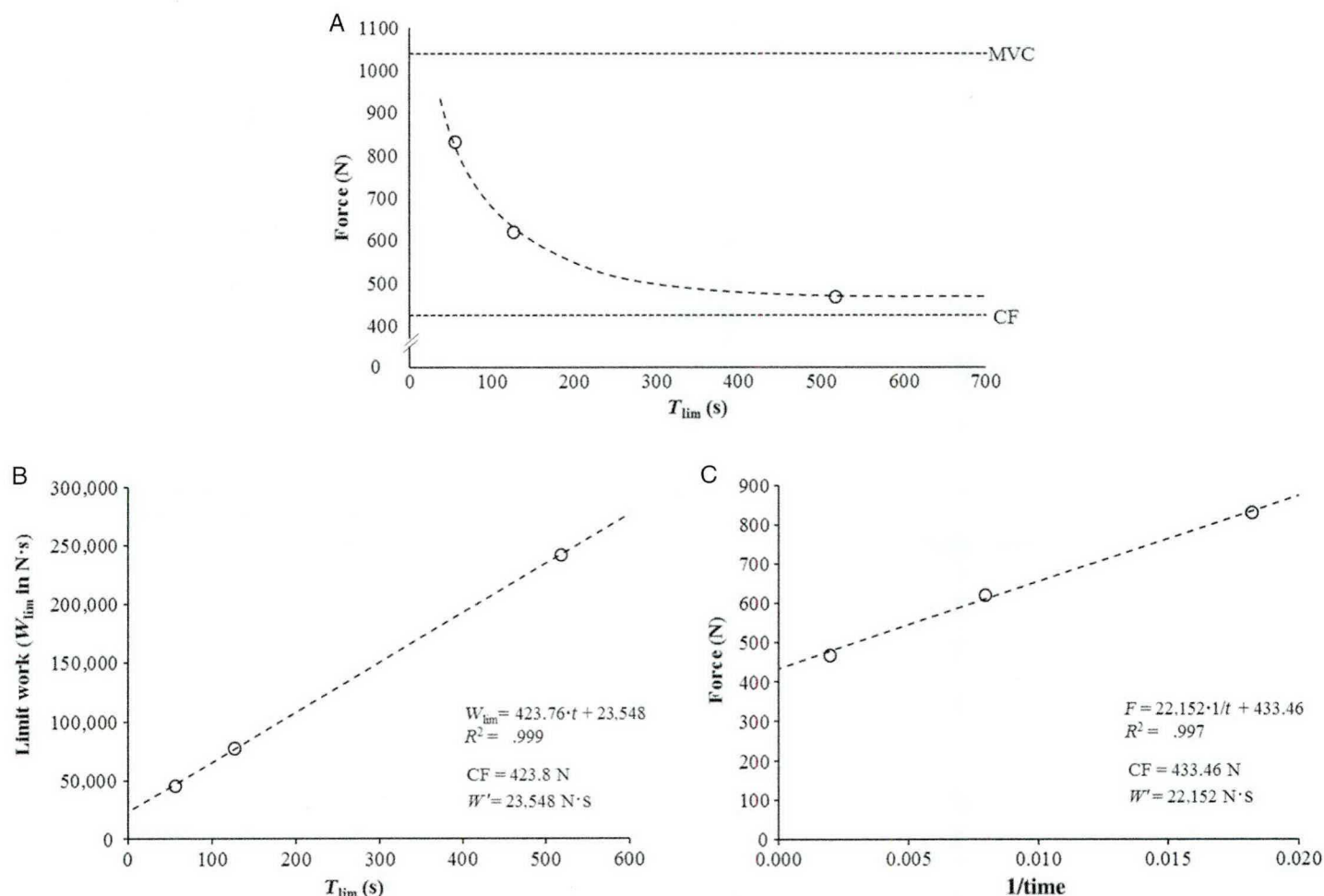
Mean CF was 425.7 (82.8) N (range: 260.1–486.4 N) and  $W'$  was 30,882 (11,820) N·s (range: 13,824–54,795 N·s) when the results of the multisession trials were modeled using the work-time model (Table 1). CF expressed as a percentage of MVC was 41.0% (6.2%). All subjects continued to task failure at 80% and 60% MVC (mean  $T_{lim}$  at 80%: 75 [29] s; at 60%: 235 [150] s); however, 3 of the 11 subjects did not reach volitional fatigue at 45% ( $T_{lim} = 626$  [360] s). The 3 subjects were invited to complete a fourth trial at 55% MVC, 2 reached task failure at this percentage, with CF of 53.5% and 48.0% MVC; the third (subject 9; Table 1) did not reach task failure within 20 minutes, suggesting a CF lying between 55% and 60% MVC.

### Comparison of Single-Session and Multisession 3-Parameter Protocols

Figure 4 illustrates the relationship and bias  $\pm 95\%$  limits of agreement between the multisession and single-session 3-parameter CF protocols. The 3 subjects who did not reach task failure at 45% MVC were excluded from analysis. The intraclass correlation coefficient between the multisession and single-session protocol for CF was .900 (95% confidence interval, .616–.979) and for  $W'$  was .768 (95% confidence interval, .190–.949).

## Discussion

The present study demonstrates that fatiguing intermittent isometric muscular contractions of the finger flexors to volitional exhaustion at multiple intensities results in the same type of relationship previously demonstrated for other synergistic muscle groups.<sup>9,10</sup> Force plotted against  $T_{lim}$  displayed a hyperbolic relationship, and



**Figure 3** — An example of the hyperbolic relationship between the force and time to task failure (A), and the CF and the curvature constant ( $W'$ ) estimates from the linear work–time (B) and the  $1/time$  (C) CF models of a representative subject. CF indicates critical force;  $W'$ , energy store component; MVC, maximum voluntary contraction;  $T_{lim}$ , time limit.

**Table 1 Two-Arm Isometric Finger-Flexor Strength and Parameters of the Force–Duration Relationship Derived From the 3-Parameter Multisession Test Using the Work–Time Model**

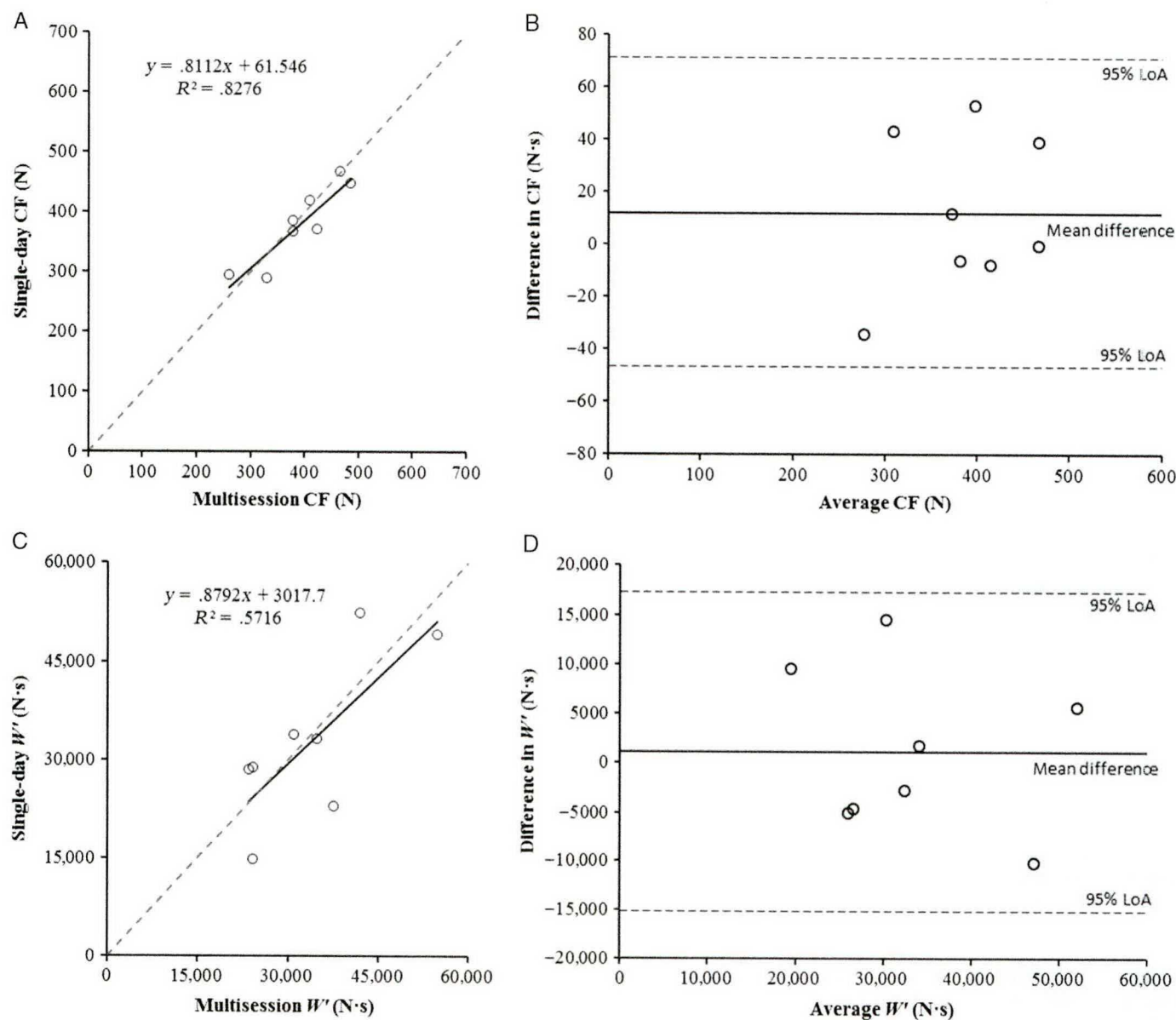
| Subject         | Isometric finger-flexor strength (MVC) |       | CF and $W'$ (work–time model) |      |            |       |
|-----------------|----------------------------------------|-------|-------------------------------|------|------------|-------|
|                 | N                                      | %BM   | CF, N                         | %MVC | $W'$ , N·s | $R^2$ |
| 1 <sup>a</sup>  | 898.8                                  | 121.7 | 480.8                         | 53.5 | 22,585     | 1.00  |
| 2               | 814.2                                  | 128.5 | 260.1                         | 31.9 | 24,275     | .94   |
| 3               | 949.1                                  | 131.3 | 331.2                         | 34.9 | 24,284     | .98   |
| 4               | 945.4                                  | 136.5 | 378.8                         | 40.1 | 31,038     | 1.00  |
| 5               | 946.7                                  | 146.9 | 378.5                         | 40.0 | 37,527     | 1.00  |
| 6               | 1093.8                                 | 147.3 | 466.1                         | 42.6 | 34,916     | 1.00  |
| 7               | 1039.9                                 | 155.2 | 423.8                         | 40.8 | 23,548     | 1.00  |
| 8               | 1091.4                                 | 156.7 | 410.9                         | 37.6 | 54,795     | .94   |
| 9 <sup>b</sup>  | 949.1                                  | 160.2 |                               |      |            |       |
| 10 <sup>a</sup> | 1126.9                                 | 166.5 | 541.0                         | 48.0 | 13,824     | 1.00  |
| 11              | 1199.3                                 | 183.0 | 486.4                         | 40.6 | 42,028     | .99   |
| Mean            | 1005.0                                 | 148.5 | 415.7                         | 41.0 | 30,882     | .99   |
| SD              | 114.0                                  | 18.2  | 82.8                          | 6.2  | 11,820     | .02   |

Abbreviations: BM, body mass; CF, critical force; MVC, maximum voluntary contraction; N·s, newton seconds;  $R$ , correlation coefficient;  $W'$ , energy store component.

<sup>a</sup> Subject did not reach task failure at 45% MVC; was invited to perform a fourth trial at 55% MVC; these values were calculated from time to failure in this additional trial.

<sup>b</sup> Subject did not reach task failure at 45% MVC, was invited to perform a fourth trial at 55% MVC and did not reach task failure.





**Figure 4** — Bland-Altman plots of the relationship (A and C) and LoA (B and D) between the multisession and single-session 3-parameter CF protocols for CF (A and B) and  $W''$  (C and D). In graphs (B) and (D), the solid horizontal line represents the mean difference between tests, and the dashed lines represent upper and lower 95% LoA. CF indicates critical force; LoA, limits of agreement.

correlation coefficients of the parameter estimates from both work-time and 1/time models were consistently very high. The observed relationships of the present study are comparable with previous research that has also shown a linear relationship between  $W_{lim}$  and  $T_{lim}$  for continuous isometric muscle contractions of the forearm flexors and extensors<sup>9</sup> and intermittent isometric finger flexor dynamometry.<sup>10</sup> The results show CF for 2-arm intermittent isometric finger flexion to be 41.0% (6.2%) of MVC for advanced to elite rock climbers. In addition, we demonstrate that a single visit to determine CF is a reliable measure that overcomes the time-consuming and potentially disruptive nature of multisession assessments.<sup>18,19</sup> However, single-session  $W''$  was less reliable. The exact reason for differences in  $W''$  remains unclear, although differences in the energy store component between ecologically valid settings and laboratory-based testing have been reported.<sup>19</sup> The results of the present study provide the first demonstration of

the sensitivity of a 3-parameter model for determining CF in the finger flexors of rock climbers. Given the use of CF in other sports, we expect it to become a common test used by coaches for understanding exercise tolerance in climbers and determining optimal training prescription. This is particularly likely given that rock climbing is now an Olympic sport.

The maximum steady state work rate that CF represents provides a useful tool for fitness diagnostics, monitoring of the physical impact of training, and a framework for exploring and understanding skeletal muscle bioenergetics and the metabolic and cardiorespiratory responses to exercise.<sup>7</sup> The present study has demonstrated the sensitivity of a simple test for the determination of CF, using equipment readily available to climbers and coaches in most climbing walls. Differences in the testing protocol, designed to maximize ecological validity in a climbing population, confounds comparison with previous data from single-arm continuous



muscle actions of the forearm flexors, middle finger flexors and finger flexors.<sup>8–10</sup> Notably, differences include (1) continuous constant power exercise bouts are unlikely to provide relevant information on climbers performance given this test to failure cannot distinguish ability groups or disciplines within the sport<sup>1,15</sup>; (2) differences in work-to-relief ratio of intermittent tests are known to alter both CP and  $W'$ <sup>20</sup>; and (3) hand and arm positions influence climbers force production, positions that involve the hand above the shoulder, and with greater than 90° extension of the elbow are known to have greater criterion validity with climbing ability than handgrip dynamometry.<sup>13</sup> Further work is necessary to establish norms of CF and  $W'$  data for comparison between ability groups and disciplines and to gain further understanding of the potential mechanisms associated with the fatigue resistance of the finger flexors of climbers.

The ability to monitor alterations in CF and  $W'$  of the finger flexors in response to interventions, including training, is likely to be invaluable. Furthermore, as the determination of CF and  $W'$  allows for the accurate prediction of  $T_{lim}$  (in seconds) at specific intermittent exercise intensities, performance capacity can be calculated. For instance, the force required to elicit a specific  $T_{lim}$  will be

$$F = W' / T_{lim} + CF \quad (3)$$

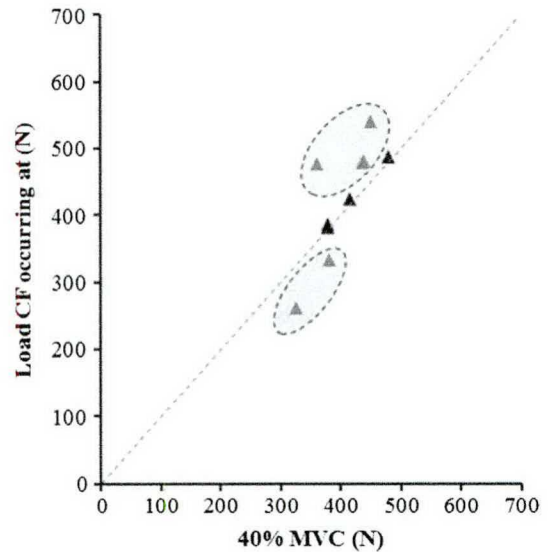
and the  $T_{lim}$  at a specified force is:

$$T_{lim} = W' / (F - CF) \quad (4)$$

If the climber's MVC is 898.8 N, CF is 480.8 N, and their  $W'$  is 22,585 N·s, the force required to elicit a  $T_{lim}$  of 120 seconds would be 669 N (74.4% MVC). Conversely,  $T_{lim}$  at 90% MVC (808.92 N) would be 68.8 seconds. It is also possible to optimize the design of interval sessions, taking into consideration the depletion of  $W'$  during the work interval and restoration of  $W'$  during the recovery interval, see Morton and Billat.<sup>21</sup> Feasibly, knowing an athlete's CF and  $W'$  would allow the coach to develop exhaustive training sessions, resulting in beneficial adaptations without overreaching.

Previously, the force–time integral (analogous to  $W_{lim}$ ) derived from a single bout of exhaustive exercise at a specific percentage of MVC has been used as a performance marker to discriminate between factors including climbing disciplines, ability, and recovery techniques.<sup>1,3,11,15</sup> The force used to determine the integral has typically been set at 40% or 60% of the climbers' finger flexor MVC. However, given that the myocellular environment and aerobic metabolism can differ dramatically between individuals working at the same percentage of MVC, the use of intensities determined in this way may have confounded the findings in previous literature.<sup>10,22</sup> This is especially true when the percentage of MVC work is completed at, for example, 40% MVC, is around the CF of the task.<sup>7</sup> This is exemplified in the results of the present study; while agreement between MVC and CF ( $P = .007$ ;  $r = .788$ ) was greater than that seen in previous studies of the forearms,<sup>10</sup> 40% of MVC would still result in some individuals performing a task <CF and others >CF (Figure 5). Therefore, the selection of exercise intensities relative to an individual's MVC could result in subjects exercising at different aerobic metabolic intensity domains, confounding the interpretation of results. Conversely, CF represents an intensity that reflects functional aerobic metabolic capacity, thus the findings of the present study, and those previously,<sup>10</sup> suggest the adoption of CF for the determination of relative exercise intensities in future research.

Protocols that require multiple visits can be time consuming and potentially disruptive to training programs.<sup>18,19</sup> Consequently, we



**Figure 5** — The relationship between the percentages of MVC that CF occurs at and 40% of MVC. Circles indicate where 40% of subjects' MVC would represent a load more than 10% greater than or less than CF. CF indicates critical force; MVC, maximum voluntary contraction.

also set out to determine the agreement of a single-session assessment of CF, thus subjects completed a single-session test of MVC as well as 3 tests at 80%, 60%, and 45% of MVC. It is well documented that CP is unaffected by prior bouts of exercises, even when exhaustive.<sup>23–26</sup> Indeed, this was found to be true in the present study, CF differed by a small margin between the single-session and multisession protocols. However, the agreement of  $W'$  was weaker, with considerably larger intraclass correlation coefficient confidence intervals. Differences between laboratory and field estimates of  $W'$  have been reported in the literature;  $W'$  (but not CP) has been found to be altered by prior high-intensity exercise, depending on the duration of subsequent recovery<sup>27</sup> and the extent of  $W'$  utilization.<sup>23</sup> It is conceivable that the 20-minute recovery period provided between tests was insufficient for complete  $W'$  reconstitution. Unlike CF, the effect of prior exercise on  $W'$  appears to be related to the duration and intensity of preceding exercise and the duration of recovery between exercise bouts.<sup>24,25,27</sup> Ferguson et al<sup>25</sup> showed that  $W'$  recovered to 86% after 15 minutes of recovery following exhaustive constant work rate exercise. Similarly, Burnley et al<sup>24</sup> and Vanhatalo and Jones<sup>27</sup> showed recovery to nonexhaustive exercise to occur in ~10 minutes. Consequently, 20 minutes was chosen as a practical duration allowing for recovery, while minimizing the time for the athlete to cooldown. However, given the error in  $W'$ , it is possible that recovery time was inadequate. One potential theory is that the reconstitution of  $W'$  is exponential; Skiba et al<sup>28</sup> reported a time constant of ~377 seconds, suggesting a recovery duration of 25 minutes is required. It is also conceivable that this is related to the size of the exercising muscle(s), as the work of Skiba et al<sup>28</sup> was in running performance. Therefore, future studies should consider longer recovery durations if a single-session test is used for the assessment of CF in small muscle groups, such as the finger flexors.

This paper has made significant steps in the development of a climbing-specific finger flexor CF protocol; however, a number of limitations should be acknowledged. (1) Both arms' finger flexors were tested together; pilot work identified that the greater amount of weight required for a single-arm test increased the resistance of



the pulleys used to adjust the load by an unacceptable amount. Given the difference in finger flexor oxygen kinetics between the dominant and nondominant arms identified previously,<sup>4</sup> future studies should consider the testing of single arms. (2) It was observed that subjects struggled with the perfect execution of the 7 seconds of work. Had it been measured, we would expect to observe that subjects were unable to produce a perfect square wave of force production, as has previously been reported.<sup>10</sup> There is potential for variability as a result, especially as the rate of force production has been found to differentiate climbers of different disciplines.<sup>29</sup> Familiarization and the experience of the subjects in the present study will have helped to reduce this; however, future work should consider the recording and calculation of actual work done. (3) Finally, the intermittent test against a fixed workload is not a perfect model of the changes in pace and length of isometric contractions of the finger flexors required for climbing. The sport involves frequent changes in exercise intensity, as dictated by the climbing route; performance may be better described by an intermittent model that takes into consideration both intense bouts of exercise and periods of rest and lower intensity exercise (see Jones and Vanhatalo<sup>30</sup> for a recent review).

## Practical Applications

The present study has demonstrated the sensitivity of a simple test for the determination of CF, using equipment readily available to climbers and coaches in most climbing gyms. Although further work is still necessary, the test of CF described is of value to coaches and climbers for understanding exercise tolerance and determine optimal training prescription to monitor improvements in the performance of the forearm flexors. Furthermore, it is recommended that future climbing studies should adopt CF for the determination of relative exercise intensities.

## Conclusions

The performance of fatiguing intermittent isometric contractions of the finger flexors at 3 percentages of climbers MVC resulted in  $T_{lim}$  and force data with very high correlation coefficients for both work–time and 1/time CF models. The observed relationships are comparable with previous research that has also shown a linear relationship between force and  $T_{lim}$  for continuous isometric muscle contractions of the forearm flexors and extensors<sup>9</sup> and intermittent isometric finger flexor dynamometry.<sup>10</sup> The results show CF of the climbers for 2-arm intermittent isometric finger flexion to be 41.0% (6.2%) of MVC and  $W'$  to be 30,882 (11,820) N·s. The results demonstrate the sensitivity of the protocol for the determination of CF and  $W'$  in the finger flexors of climbers from an ecologically valid, climbing-specific multisession test and CF from a single-session test.

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